

Name: White, Rice, & Spice Team

Date: 3/9/26

Class: Design for Engineers e4usa

MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The design has to be made of high-quality 3D printed material, meaning that there cannot be any number of quality & durability fluctuations in the material.

Unlike Polylactic Acid (PLA), 3D printed material (plastic filament), does not warp. In addition, it is highly durable and has long-term sustainability. (Our prior research was on PLA and its quality, longevity, and durability.)

Our stakeholder (Dr. Sbar) agreed with the baseline requirement of having a consistent base material in order to ensure dependable solutions.

Design Requirement #2 (HIGH Priority)

The design has to be easily attachable to the soap dispenser apparatus/frame, meaning it is convenient to install, and it fits perfectly/is measured properly to adhere to the soap dispenser frame.

The few existing soap dispenser soap stoppers have a difficult installation. Prior research shows that mechanical fasteners have a snap fit, with very limited movement.

There was concern expressed about how this would be measured, in which the team came up with a satisfaction rate in installation, or reducing the steps to install/making installation as simple as possible.

Design Requirement #3 (HIGH Priority)

The design has to collect all (residual) soap & prevent it from touching the floor or dropping onto a hard-to-reach surface.

When soap is dropped on an inconvenient surface, it may stain or be a potential slip hazard, given how its disposal is in an inconvenient or inaccessible location. By containing

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all of the soap, we mitigate these risks. (Prior research shows that residual soap can pose a hazard, with contamination/germs as well as being a slip hazard.)

The team's stakeholders approved of the design requirement, citing that this design requirement was the main objective of the solution, and thus should be held at the highest priority.

Design Requirement #4 (HIGH Priority)

The design must have a proper fit within the geometry to accommodate a soap bottle and sink. The design should be a slim design that accommodates the soap bottle & sink without being obtrusive, bulky, or inconvenient/in the way.

The solution must have proper measurements & design to accommodate stakeholder needs. (Prior research shows the effectiveness & usage of a solution decreasing as stakeholder satisfaction with it decreases. If a stakeholder is unsatisfied or unhappy with the solution, it is less likely to fulfill its objective.)

Our expert evaluator approved of the design requirement as a proper fit in the design is necessary for applying our solution to stakeholders.

Design Requirement #5 (HIGH Priority)

The design must be able to hold the weight of the soap bottle's entire contents.

Prior research & evidence show a design's ineffectiveness or inconvenience through structural durability issues, as well as issues in longevity, which hinder its ability to successfully solve the problem.

The stakeholder identified and highlighted the specific aspect of the solution needing to be able to contain all the excess soap in order to fulfill the primary aspect of the solution.

Design Requirement #6 (LOW Priority)

The design does not have sharp edges to prevent injury.

Prior research suggests that sharp corners of everyday apparatuses lead to many minor injuries. By getting rid of sharp edges, either through modifying the 3D design or through covering it with soft material, users will deem the solution safe to use and use it for its intended purpose instead of disposing of it for safety reasons.

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Side note given by stakeholders just as a warning in part with dealing with user efficiency.

Design Requirement #7 (LOW Priority)

The design must be visually appealing and simple enough to encourage stakeholders to use the design. (Can be measured by how many stakeholders are attracted to use the product).

Prior research shows that, given how stakeholder usage is a necessity for the prototype to work, the prototype is more likely to be efficient if the stakeholders believe the design is not dirty or an inconvenience to use.

Stakeholder commented that #7 was not completely necessary, but would be a good addition as some might be interested in aesthetics.